

Module 1: Data Mining with Python

Description:

The purpose of the course is to provide the fundamental of Python programming. We will learn the following:

- Python fundamentals like syntax, data types, operators, conditional statements. We will also learn how to write simple python scripts.
- We will learn about objects, class and methods. We will also be using with different data types and how we can manipulate them.
- We will also learn about the Numpy library which will be used for the basic calculation.
- And also, we will learn about the Pandas library.
- Once, we get an understanding of python scripting, we will see how to connect to the database and also about the data visualization.

Course Pre-requisites:

This course is designed for the beginner level; hence, all the data science aspirants can learn this course.

The candidate should have the basic Logical skill.

Content:

Chapter 1: Introduction to Python

- Basics of Python programming language
- Built-in Data Types
- Operators and operator Precedence
- Type casting

Chapter 2: Fundamentals of Python

- Packages, modules and library
- Input and output statements
- Command line arguments
- Conditional Statements

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- Iterative Statements and Control Statements
- Data Structures
- Deep copy and Shallow copy
- String Operations
- Format Specifiers
- Iterators and Generators
- Decorators
- Pickling and Unpickling
- File handling and Exception handling
- Regular Expressions
- Date and time

Chapter 3: Object Oriented Programming's (OOPs)

- About OOPs
- Advantages of OOPs
- Class and Object
- Features of OOPs
- Encapsulation
- Polymorphism
- Inheritance
- Abstraction

Chapter 4: Numpy

- About Numpy
- Advantages over List
- Array Attributes
- Constructing Array
- Indexing and Slicing
- Data types
- Numpy Operations

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Chapter 5: Pandas

- Creating Data Frame
- Reading the data from files
- Attributes of DataFrame
- Basic operations with data frame
- Indexing and slicing
- Sub-setting data using criteria
- Sorting the values
- Dealing with null values
- Data manipulation Operations
- Date and time with Pandas

Chapter 6: Connecting database with Python

- Introduction to Database
- Types of Databases
- Types of Database Management Systems
- Constraints
- Commands in MySQL
- Clauses
- Functions
- Joins

Chapter 7: Pandasql – Run SQL Queries in Python

- SQL commands
- SQL queries for different usecases
- Introduction to pandasql
- Pandasql queries for Creating and manipulating data
- Working with Dataframes using sql queries
- Data filtering , Aggregation
- Data Joining using SQL joins

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Chapter 8: Data Visualization

- Introduction to Visualization
- Importance of Visualization
- Types of plots available
- Difference between different visualization libraries
- Difference between Histogram and Bar plot
- Basic Plotting Operations
- Introduction to Plotly and Cufflinks
- Data Visualization using Plotly
- Data Visualization using Cufflinks
- Using of Folium for Creating and visualizing locations in Map



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Module 2: Statistics with Python

Description:

The purpose of the course is to provide the basic of statistics. We will learn the following:

- We will learn about measure of central tendency and measures of variance.
- We will learn about outliers and covariance.
- We will learn about different distributions like normal, binomial, Uniform, Poison, Exponential and Poisson distribution.
- We also go through different hypothesis tests like null hypothesis one tailed and two tailed test, chi-square test. We will also understand type I and type II error.
- We will also different parametric test and non-parametric test.
- We will also learn about correlation and regression.

Course Pre-requisites:

This course is designed for the intermediate level; hence, all the data science aspirants can learn this course. The person should be good with python programming language.

The candidate should have the basic mathematic skill.

Content:

Chapter 1: Introduction to Statistics

- Introduction to statistics
- Types of data
- Population and sampling
- Sampling Methods/Techniques
- Types of Statistics
- Measure of Central Tendency
- Measure of Variance
- Outliners
- Covariance

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Chapter 2: Distributions

- About Distribution
- Types of distribution
 - Normal distribution
 - Binomial distribution
 - Uniform distribution
 - Bernoulli's distribution
 - Poisson distribution
 - Exponential distribution
- Relation between the distributions

Chapter 3: Hypothesis and computational methods

- Null Hypothesis, p-value
- One tailed and two tailed test
- Type I and type II errors
- Parametric tests
 - T-test
 - ANOVA test
 - Pearson's R test
- Non-Parametric test
 - Mann-Whitney U test
 - Wilcoxon Signed Rank test
 - Krushal-Wallis H test
 - Friedman test
 - Spearson's test (Rho)
- Chi- Square test

Chapter 4: Correlation and regression

- Introduction to regression
- Type of regression
- Co-relation
- Working with Python

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Module 3: Machine Learning

Description:

The purpose of the course is to provide the basics of Machine Learning. We will learn the following:

- Machine learning and what are the different algorithms.
- Types - supervised learning, unsupervised learning and reinforcement learning.
- Data Pre-processing and data cleaning with some examples.
- At the end, we also train one Model and we also work with a case study.

Course Pre-requisites:

This course is designed for the intermediate level; hence, all the data science aspirants can learn this course. The person should have the knowledge of python programming language and should be good in statistics.

The candidate should have the basic mathematic skill and statistical skill

Content:

Chapter 1: Introduction to ML

- About Machine Learning
- Supervised and Unsupervised learning
- Semi-supervised learning
- Reinforcement learning

Chapter 2: Data Pre-processing in ML

- Data processing
- Data cleaning
- Data transformation and reduction
- Step in pre-processing

Chapter 3: Building and training ML Models

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Chapter 4: ML Algorithms

- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machines (SVM)
- Naïve-Bayes Classifier (NB)
- K-nearest Neighbors
- K – Mean Clustering
- Apriori
- Principal Component Analysis (PCA)
- Boosting Algorithm
- Feature selection



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Module 4: Artificial Intelligence

Description:

The purpose of the course is to provide the basic of AI. We will learn the following:

- About the AI and DL. We also learn different techniques of AI.
- Neural networks and different functions from NLTK library.

Course Pre-requisites:

This course is designed for the intermediate level; hence, all the data science aspirants can learn this course. The person should be good with python programming and should have logical thinking.

The candidate should have the basic mathematic and statistical skill.

Content:

Chapter 1: Introduction to AI and DL

- Artificial Intelligence
- Deep learning
- AI v/s ML v/s DL
- DL Framework
- Types of learning in DL

Chapter 2: Optimizer, Loss and Activation

- Optimizer
- Types of optimizers
- Loss function
- Types of loss function
- Activation function
- Types of Activation function

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Chapter 3: Neural Networks

- Neuron
- Weight
- Forward Propagation
- Gradient Descent Function
- Backward Propagation
- Pooling and Flatten

Chapter 4: Introduction to NLP

- Natural Language Processing
- Beginning of NLP
- Finding its footing
- Rise of NLP
- Syntactic and Semantic Analysis

Chapter 5: NLP techniques

- Sentiment Analysis
- Named Entity Recognition
- Summarization
- Topic modeling
- Text classification
- Keyword Extraction
- Lemmatization and stemming

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